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DISLOCATIONS IN GAA TRANSISTORS AND THE METHODS OF FORMING THE SAME

Abstract

A method includes forming a protruding feature. The protruding feature includes a first sacrificial nanosheet over a bulk semiconductor substrate, a first semiconductor nanosheet over the first sacrificial nanosheet, a second sacrificial nanosheet over the first semiconductor nanosheet, and a second semiconductor nanosheet over the second sacrificial nanosheet. The method further includes forming a dummy gate stack on the protruding feature, etching the protruding feature to form a recess, forming a source/drain region in the recess, wherein dislocations are formed in the source/drain region, removing the first sacrificial nanosheet and the second sacrificial nanosheet, and forming a replacement gate stack to replace the dummy gate stack.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application claims the benefit of the following provisionally filed U.S. Patent application: Application No. 63/558,961, filed on Feb. 28, 2024, and entitled “DISLOCATION IN S/D EPI IN GAAFET,” which application is hereby incorporated herein by reference.

BACKGROUND

[0002] Technological advances in Integrated Circuit (IC) materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generations. In the course of IC evolution, functional density (for example, the number of interconnected devices per chip area) has generally increased while geometry sizes have decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

[0003] Such scaling down has also increased the complexity of processing and manufacturing ICs, and for these advances to be realized, similar developments in IC processing and manufacturing are needed. For example, Gate-All-Around (GAA) Transistors have been introduced to replace planar transistors. The structures of the GAA transistors and methods of fabricating the GAA transistors are being developed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. 1-4, 5A, 5B, 6A, 6B, 7A, 7B, 8A, 8B, 9A, 9B, 10A, 10B, 11A, 11B, 12A, 12B, 13A, 13B, 14A, and 14B illustrate the views of intermediate stages in the formation of a Gate All-Around (GAA) transistor in accordance with some embodiments.

[0006] FIGS. 15 through FIGS. 20A and 20B illustrate the cross-sectional views of intermediate stages in the formation of source/drain regions and some overlying features in accordance with some embodiments.

[0007] FIG. 21 illustrates a process flow for forming dielectric regions and overlying source/drain regions in accordance with some embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which

additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] A Gate All-Around (GAA) transistor that comprises source/drain regions having dislocations is provided. In accordance with some embodiments, the processes for forming the source/drain regions are adjusted, so that dislocations are formed in the source/drain regions. Although GAA transistors are used as an example to discuss the concept of the present disclosure, the embodiments may be applied on other types of transistors including and not limited to Fin Field-Effect Transistors (FinFETs), planar transistors, and the like. Embodiments discussed herein are to provide examples to enable making or using the subject matter of this disclosure, and a person having ordinary skill in the art will readily understand modifications that can be made while remaining within contemplated scopes of different embodiments. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. Although method embodiments may be discussed as being performed in a particular order, other method embodiments may be performed in any logical order.

[0011] FIGS. **1-4**, **5A**, **5B**, **6A**, **6B**, **7A**, **7B**, **8A**, **8B**, **9A**, **9B**, **10A**, **10B**, **11A**, **11B**, **12A**, **12B**, **13A**, **13B**, **14A**, and **14B** illustrate the cross-sectional views of intermediate stages in the formation of a GAA transistor in accordance with some embodiments of the present disclosure. The corresponding processes are also reflected schematically in the process flow shown in FIG. **21**.

[0012] Referring to FIG. **1**, a perspective view of wafer **10** is shown. Wafer **10** includes a multilayer structure comprising multilayer stack **22** on substrate **20**. In accordance with some embodiments, substrate **20** is a semiconductor substrate, which may be a silicon substrate, a silicon germanium (SiGe) substrate, or the like, while other substrates and/or structures, such as semiconductor-on-insulator (SOI), strained SOI, silicon germanium on insulator, or the like, could be used. Substrate **20** may be doped as a p-type semiconductor, although in other embodiments, it may be doped as an n-type semiconductor.

[0013] In accordance with some embodiments, multilayer stack **22** is formed through a series of deposition processes for depositing alternating materials. The respective process is illustrated as process **202** in the process flow **200** shown in FIG. **21**. In accordance with some embodiments, multilayer stack **22** comprises first layers **22A** formed of a first semiconductor material and second layers **22B** formed of a second semiconductor material different from the first semiconductor material.

[0014] In accordance with some embodiments, the first semiconductor material of a first layer **22A** is formed of or comprises SiGe, Ge, Si, GaAs, InSb, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, or the like. In accordance with some embodiments, the deposition of first layers **22A** (for example, SiGe) is through epitaxial growth, and the corresponding deposition method may be Vapor-Phase Epitaxy (VPE), Molecular Beam Epitaxy (MBE), Chemical Vapor deposition (CVD), Low Pressure CVD (LPCVD), Atomic Layer Deposition (ALD), Ultra High Vacuum CVD (UHVCVD), Reduced Pressure CVD (RPCVD), or the like. In accordance with some embodiments, the first layer **22A** is formed to a first thickness in the range between about 30 Å and about 300 Å. However, any suitable thickness may be utilized while remaining within the scope of the embodiments.

[0015] Once the first layer **22A** has been deposited over substrate **20**, a second layer **22B** is deposited over the first layer **22A**. In accordance with some embodiments, the second layers **22B** is formed of or comprises a second semiconductor material such as Si, SiGe, Ge, GaAs, InSb, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, combinations of these, or the like, with the second semiconductor material being different from the first semiconductor material of first layer **22A**. For example, in accordance with some embodiments in which the first layer **22A** is silicon germanium, the second layer **22B** may be formed of silicon, or vice versa. It is appreciated that any suitable combination of materials may be utilized for first layers **22A** and the second layers **22B**.

[0016] In accordance with some embodiments, the second layer **22B** is epitaxially grown on the first layer **22A** using a deposition technique similar to that is used to form the first layer **22A**. In accordance with some embodiments, the second layer **22B** is formed to a similar thickness to that of the first layer **22A**. The second layer **22B** may also be formed to a thickness that is different from the first layer **22A**. In accordance with some embodiments, the second layer **22A** has thickness in the range between about 4 nm and 7 nm, while the second layer **22B** has thickness in the range between about 8 nm and 12 nm, for example.

[0017] Once the second layer **22B** has been formed over the first layer **22A**, the deposition process is repeated to form the remaining layers in multilayer stack **22**, until a desired topmost layer of multilayer stack **22** has been formed. In accordance with some embodiments, first layers **22A** have thicknesses the same as or similar to each other, and second layers **22B** have thicknesses the same as or similar to each other. First layers **22A** may also have the same thicknesses as, or different thicknesses from, that of second layers **22B**. In accordance with some embodiments, first layers **22A** are removed in the subsequent processes, and are alternatively referred to as sacrificial layers **22A** throughout the description. In accordance with alternative embodiments, second layers **22B** are sacrificial, and are removed in the subsequent processes.

[0018] In accordance with some embodiments, there may be some pad oxide layer(s) and hard mask layer(s) (not shown) formed over multilayer stack **22**. These layers are patterned, and are used for the subsequent patterning of multilayer stack **22**.

[0019] Referring to FIG. 2, multilayer stack **22** and a portion of the underlying substrate **20** are patterned in an etching process(es), so that trenches **23** are formed. The respective process is illustrated as process **204** in the process flow **200** shown in FIG. 21. Trenches **23** extend into substrate **20**. The remaining portions of multilayer stacks are referred to as multilayer stacks **22'** hereinafter. Underlying multilayer stacks **22'**, some portions of substrate **20** are left, and are referred to as substrate strips **20'** hereinafter. Multilayer stacks **22'** include semiconductor layers **22A** and **22B**. Semiconductor layers **22A** are alternatively referred to as sacrificial layers, and Semiconductor layers **22B** are alternatively referred to as nanostructures hereinafter. The portions of multilayer stacks **22'** and the underlying substrate strips **20'** are collectively referred to as semiconductor strips **24**.

[0020] In above-illustrated embodiments, the GAA transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0021] FIG. 3 illustrates the formation of isolation regions **26**, which are also referred to as Shallow Trench Isolation (STI) regions throughout the description. The respective process is illustrated as process **206** in the process flow **200** shown in FIG. 21. STI regions **26** may include a liner oxide (not shown), which may be a thermal oxide formed through the thermal oxidation of a surface layer

of substrate **20**. The liner oxide may also be a deposited silicon oxide layer formed using, for example, ALD, High-Density Plasma Chemical Vapor Deposition (HDPCVD), CVD, or the like. STI regions **26** may also include a dielectric material over the liner oxide, wherein the dielectric material may be formed using Flowable Chemical Vapor Deposition (FCVD), spin-on coating, HDPCVD, or the like. A planarization process such as a Chemical Mechanical Polish (CMP) process or a mechanical grinding process may then be performed to level the top surface of the dielectric material, and the remaining portions of the dielectric material are STI regions **26**. [0022] STI regions **26** are then recessed, so that the top portions of semiconductor strips **24** protrude higher than the top surfaces **26T** of the remaining portions of STI regions **26** to form protruding fins **28**. Protruding fins **28** include multilayer stacks **22'** and the top portions of substrate strips **20'**. The recessing of STI regions **26** may be performed through a dry etching process, wherein NF₃ and NH₃, for example, are used as the etching gases. During the etching process, plasma may be generated. Argon may also be included. In accordance with alternative embodiments of the present disclosure, the recessing of STI regions **26** is performed through a wet etching process. The etching chemical may include HF, for example.

[0023] Referring to FIG. 4, dummy gate stacks **30** and gate spacers **38** are formed on the top surfaces and the sidewalls of (protruding) fins **28**. The respective process is illustrated as process **208** in the process flow **200** shown in FIG. 21. Dummy gate stacks **30** may include dummy gate dielectrics **32** and dummy gate electrodes **34** over dummy gate dielectrics **32**. Dummy gate dielectrics **32** may be formed by oxidizing the surface portions of protruding fins **28** to form oxide layers, or by depositing a dielectric layer such as a silicon oxide layer. Dummy gate electrodes **34** may be formed, for example, using polysilicon or amorphous silicon, and other materials such as amorphous carbon may also be used.

[0024] Each of dummy gate stacks **30** may also include one (or a plurality of) hard mask layer **36** over dummy gate electrode **34**. Hard mask layers **36** may be formed of silicon nitride, silicon oxide, silicon carbo-nitride, silicon oxy-carbo nitride, or multilayers thereof. Dummy gate stacks **30** may cross over a single one or a plurality of protruding fins **28** and the STI regions **26** between protruding fins **28**. Dummy gate stacks **30** also have lengthwise directions perpendicular to the lengthwise directions of protruding fins **28**. The formation of dummy gate stacks **30** includes forming a dummy gate dielectric layer, depositing a dummy gate electrode layer over the dummy gate dielectric layer, depositing one or more hard mask layers, and then patterning the formed layers through a patterning process(es).

[0025] Next, gate spacers **38** are formed on the sidewalls of dummy gate stacks **30**. In accordance with some embodiments of the present disclosure, gate spacers **38** are formed of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO), silicon carbide (SiC), silicon oxide (SiO₂), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like, and may have a single-layer structure or a multilayer structure including a plurality of dielectric layers. The formation process of gate spacers **38** may include depositing one or a plurality of dielectric layers, and then performing an anisotropic etching process(es) on the dielectric layer(s). The remaining portions of the dielectric layer(s) are gate spacers **38**.

[0026] FIGS. 5A and 5B illustrate the cross-sectional views of the structure shown in FIG. 4. FIG. 5A illustrates the reference cross-section A1-A1 in FIG. 4, which cross-section cuts through the portions of protruding fins **28** not covered by gate stacks **30** and gate spacers **38**, and is perpendicular to the gate-length direction. FIG. 5B illustrates the reference cross-section B-B in FIG. 4, which reference cross-section is parallel to the lengthwise directions of protruding fins **28**.

[0027] Referring to FIGS. 6A and 6B, the portions of protruding fins **28** (FIG. 4) that are not directly underlying dummy gate stacks **30** and gate spacers **38** are recessed through an etching process to form recesses **42**. The respective process is illustrated as process **210** in the process flow **200** shown in FIG. 21. For example, a dry etch process may be performed using C.sub.2F.sub.6, CF.sub.4, SO.sub.2, the mixture of HBr, Cl.sub.2, and O.sub.2, the mixture of HBr, Cl.sub.2,

O.sub.2, and CH.sub.2F.sub.2, or the like to etch multilayer semiconductor stacks 22' and the underlying substrate strips 20'. The bottoms of recesses 42 are at least level with, or may be lower than (as shown in FIG. 6B), the bottoms of multilayer semiconductor stacks 22'. The etching may be anisotropic, so that the sidewalls of multilayer semiconductor stacks 22' facing recesses 42 are vertical and straight, as shown in FIG. 6B.

[0028] Referring to FIGS. 7A and 7B, sacrificial semiconductor layers 22A are laterally recessed to form lateral recesses 41, which are recessed from the edges of the respective overlying and underlying nanostructures 22B. The respective process is illustrated as process 212 in the process flow 200 shown in FIG. 21. The lateral recessing of sacrificial semiconductor layers 22A may be achieved through a wet etching process using an etchant that is more selective to the material (for example, silicon germanium (SiGe)) of sacrificial semiconductor layers 22A than the material (for example, silicon (Si)) of the nanostructures 22B and substrate 20. For example, in an embodiment in which sacrificial semiconductor layers 22A are formed of silicon germanium and the nanostructures 22B are formed of silicon, the wet etching process may be performed using an etchant such as hydrochloric acid (HCl). The wet etching process may be performed using a dip process, a spray process, a spin-on process, or the like.

[0029] In accordance with alternative embodiments, the lateral recessing of sacrificial semiconductor layers 22A is performed through an isotropic dry etching process or a combination of a dry etching process and a wet etching process.

[0030] Referring to FIGS. 8A and 8B, inner spacers 44 are formed. The respective process is illustrated as process 214 in the process flow 200 shown in FIG. 21. In accordance with some embodiments, the formation of inner spacers 44 includes depositing a conformal dielectric layer, which extends into the lateral recesses 41 (FIG. 7B). Next, an etching process (also referred to as a spacer trimming process) is performed to trim the portions of the spacer layer outside of the lateral recesses 41, leaving the portions of the spacer layer in the lateral recesses 41. The remaining portions of the spacer layer are referred to as inner spacers 44.

[0031] Referring to FIGS. 9A and 9B, dielectric layers 46 may be formed before the formation of source/drain regions 48. The respective process is illustrated as process 215 in the process flow 200 shown in FIG. 21. Alternatively, dielectric layers 46 are not formed. Accordingly, dielectric layers 46 are illustrated as being dashed to indicate that dielectric layers 46 may be formed or not formed. Next, epitaxial source/drain regions 48 and dislocations 49 (referring to FIG. 17) are formed in recesses 42. The respective process is illustrated as process 216 in the process flow 200 shown in FIG. 21. The details of source/drain regions 48 and dislocations 49 are illustrated in FIG. 17.

[0032] FIGS. 15-17 illustrate the details in the formation of the dielectric layers and source/drain regions (as shown in FIGS. 9A and 9B) in accordance with some embodiments. FIG. 15 illustrates the region 45 in FIG. 8B, in which recesses 42 and inner spacers 44 have been formed. Next, dielectric layer 46 is formed at the bottom of recess 42 in accordance with some embodiments. Dielectric layer 46 may comprise silicon nitride (SiN). The process gas may include silane (SiH.sub.4), ammonia (NH.sub.3), and the like. Dielectric layer 46 may also be formed of or comprise silicon oxide (SiO), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbo-nitride (SiCN), silicon oxy carbo-nitride (SiOCN), or the like. In accordance with some embodiments, dielectric layer 46 has a multilayer structure, for example, including a conformal silicon oxide liner and a silicon nitride region on the silicon oxide liner.

[0033] In accordance with some embodiments, the formation of dielectric layer 46 includes a deposition process, followed by etching the sidewall portions of dielectric layer in recess 42, and the removal of the top portions over the top surfaces of dummy gate stacks 30 (also refer to FIG. 8B). Dielectric layer 46 may be deposited using a directional deposition process, which includes both of an anisotropic component and an isotropic component.

[0034] The sidewall portions of dielectric layer 46 in the recesses and on the sidewalls of the structure protruding higher than substrate 20 may be thinner than the bottom portion at the bottom

of recess **42** and the top portions over dummy gate stack **30**. An isotropic etching process is then performed to remove the thin sidewall portions. The top portions of the dielectric layer on top of dummy gate stacks **30** may be removed by using a sacrificial layer to fill recess **42** and to protect the bottom portion, and performing an etching process. In accordance with alternative embodiments, dielectric layer **46** is not formed. The semiconductor layer **46** is thus illustrated as being dashed to indicate that it may be or may not be formed.

[0035] FIGS. **16** and **17** illustrate the selective formation of epitaxy regions **48** (source/drain regions) in accordance with some embodiments. Source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context. FIG. **16** illustrates the epitaxy of semiconductor layer **48A** (also referred to as layer-1 or L1) through a selective epitaxy growth process. The resulting semiconductor layer **48A** is selectively grown from the sidewalls of nanostructures **22B**. When there is no dielectric layer **46** formed, semiconductor layer **48A** is also grown from the exposed top surface of semiconductor substrate **20**. On the other hand, no portion of semiconductor layer **48A** is grown from dielectric features such as inner spacers **44**, gate spacers **38**, and hard mask **36** (refer to FIG. **9**).

[0036] The selective formation process may include a plurality of cycles, each including a deposition process and an etch-back process. In the deposition process, the thickness of semiconductor layer **48A** is increased, while in the etch-back process, the thickness of semiconductor layer **48A** is reduced. The plurality of cycles are also referred to as deposition- and-etch cycles.

[0037] When the source/drain region is an n-type region of an n-type transistor, semiconductor layer **48A** may comprise silicon or SiC (and may or may not comprise a small amount of germanium) and an n-type dopant such as As, P, Sb, or the like, or combinations thereof. For example, semiconductor layer **48A** may comprise SiAs, SiP, SiCP, SiAsP, SiSb, or the like. The semiconductor layer **48A** may have an n-type dopant concentration in a range between about $1\text{E}20/\text{cm}^3$ and about $2\text{E}21/\text{cm}^3$. The thickness of semiconductor layer **48A** may be smaller than about 10 nm.

[0038] In the deposition, the process gases for forming the n-type semiconductor layer **48A** may include SiH₄, dichlorosilane (DCS), HCl, GeH₄, PH₃, or the like. The wafer temperature may be in the range between about 500° C. and about 850° C., the chamber pressure may be in the range between about 4 torr and about 300 torr.

[0039] When the source/drain region is a p-type region of a p-type transistor, semiconductor layer **48A** may comprise silicon, SiGe, or Ge, and further includes a p-type dopant such as boron, indium, or combinations thereof. For example, semiconductor layer **48A** may comprise SiGeB, GeB, or the like. The semiconductor layer **48A** may have a p-type dopant concentration in a range lower than about $5\text{E}20/\text{cm}^3$, such as in a range between about $1\text{E}20/\text{cm}^3$ and about $5\text{E}20/\text{cm}^3$. The thickness of semiconductor layer **48A** may be smaller than about 10 nm.

[0040] In the deposition of the p-type semiconductor layer **48A**, the process gases may include SiH₄, DCS, HCl, GeH₄, BH₃, BCl₃, or the like. The wafer temperature may be in the range between about 400° C. and about 850° C., the chamber pressure may be in the range between about 4 torr and about 300 torr.

[0041] In accordance with some embodiments, as discussed in the preceding paragraphs, the process gases may include an etching gas for etching semiconductor layer **48A**. For example, the etching gas may include HCl. The etching gas helps to remove the deposited semiconductor layer **48A** on dielectric features such as inner spacers **44**, gate spacers **38**, hard masks **36**, and dielectric layer **46**, if any. In order to form high-quality semiconductor layer **48A**, the deposition rate of semiconductor layer **48A** is kept low, for example, lower than about 10 Å/minute.

[0042] After the deposition of one layer of semiconductor layer **48A**, an etch-back process is performed to etch-back semiconductor layer **48A**. This helps to remove any deposited semiconductor on dielectric features. In accordance with some embodiments, if any dislocation is

formed in semiconductor layer **48A**, the portions of semiconductor layer **48A** having the dislocations are etched. Accordingly, the semiconductor layer **48A** remaining after the etch-back process does not include dislocations therein. As a result of the etch-back process and the low growth rate, semiconductor layer **48A** may be free from dislocations therein, or may have a low number of dislocations.

[0043] FIG. **17** illustrates the epitaxy growth of semiconductor layer **48B** (also referred to as semiconductor layer-2 or L2) and the dislocations **49** therein. The resulting semiconductor layer **48B** is grown from, and is different from semiconductor layer **48A**. For example, semiconductor layer **48B** may be deposited as having an element (such as arsenic (As)) that is not in semiconductor layer **48A**, or vice versa. Semiconductor layers **48A** and **48B** may have a same element (such as Si) but have different percentages of the element(s). The deposition process may be selective, so that no semiconductor layer **48B** is directly grown from dielectric features such as inner spacers **44**, gate spacers **38**, and hard mask **36**.

[0044] It is appreciated, however, as shown in FIG. **16**, some dielectric features such as inner spacers **44** and dielectric layer **46** may still have surfaces exposed to recess **42**, and semiconductor layer **48B** may grow from semiconductor layer **48A** and grown to the dielectric features, so that semiconductor layer **48B** may be in contact with the dielectric features. When the deposition of semiconductor layer **48B** is ended, the top surface of semiconductor layer **48B** may be higher than the top surface of the topmost nanostructure **22B**.

[0045] When the source/drain region **48** is an n-type region of an n-type transistor, semiconductor layer **48B** may comprise silicon or SiC (and may or may not comprises a small amount of germanium) and an n-type dopant such as phosphorous. For example, semiconductor layer **48B** may comprise, SiP, SiCP, or the like, and may be free from n-type dopants such as As, Sb, and the like (which are doped in semiconductor layer **48B**). The semiconductor layer **48B** may have an n-type dopant concentration higher than the n-type dopant concentration in semiconductor layer **48A**. For example, the n-type dopant concentration in semiconductor layer **48B** may be in a range between about $5E20/\text{cm}^3$ and about $5E21/\text{cm}^3$.

[0046] In the deposition of the n-type semiconductor layer **48B**, the process gases may include SiH₄, DCS, HCl, GeH₄, PH₃, or the like. The wafer temperature may be in the range between about 500° C. and about 850° C., the chamber pressure may be in the range between about 4 torr and about 300 torr.

[0047] When the source/drain region is a p-type region of a p-type transistor, semiconductor layer **48B** may comprise SiGe or Ge, and may further include a p-type dopant such as boron, indium, or combinations thereof. For example, semiconductor layer **48B** may comprise SiGeB, GeB, or the like. The germanium atomic percentage may be greater than the germanium atomic percentage in semiconductor layer **48A**, for example, with a difference being greater than about 20 percent or 30 percent. For example, the germanium atomic percentage in semiconductor layer **48B** may be in the range between about 50 percent and about 60 percent. The p-type dopant concentration in semiconductor layer **48B** may be higher than the p-type dopant concentration in semiconductor layer **48A**. For example, the p-type dopant concentration in semiconductor layer **48B** may be in a range between about $7E20/\text{cm}^3$ and about $1E21/\text{cm}^3$.

[0048] In the deposition of the p-type semiconductor layer **48B**, the process gases may include SiH₄, DCS, HCl, GeH₄, BH₃, BCl₃, or the like. The wafer temperature may be in the range between about 400° C. and about 850° C., the chamber pressure may be in the range between about 4 torr and about 300 torr.

[0049] In accordance with some embodiments, the deposition process gas may further include an etching gas such as HCl, so that the semiconductor layer **48B** is not grown on gate spacers **38** and hard masks **36** (FIG. **8B**).

[0050] During the formation of semiconductor layer **48B**, dislocations **49** (including dislocations **49A** and **49B**) are formed and grown along with the deposition of semiconductor layer **48B**. The

length of dislocations **49** may be in the range between about 1 nm and about 70 nm. The total count of dislocations **49** may be from 1 to several hundreds. The dislocations may include the ones with opposite slopes, for example, the ones in the bottom-left to top-right direction, and the ones in the bottom-right to top-left direction.

[0051] In order to form dislocations **49**, the formation process of semiconductor layer **48B** is adjusted to be different from that of semiconductor layer **48A**. In accordance with some embodiments, the growth of semiconductor layer **48B** is continuously performed without etch-back process therein. Alternatively stated, the formation of semiconductor layer **48B** may be a continuous growth process until the top surface of semiconductor layer **48B** is higher than the top surface of the top semiconductor nanostructure **22B**, without etch-back process therein. Accordingly, during the growth, the dislocations **49** have a chance to further grow, rather than be removed in the etch-back process. If etch-back is performed, semiconductor layer **48B** may not have dislocations therein.

[0052] It is appreciated that when the growth rate of semiconductor layer **48B** is low, dislocations may not be able to be formed, even if no etch-back is performed in the formation of semiconductor layer **48B**. To ensure the formation of dislocations **49**, the growth rate of semiconductor layer **48B** is increased. In accordance with some embodiments, the deposition rate of semiconductor layer **48B** is relatively high, for example, higher than about 20 Å/minute, and may be in the range between about 20 Å/minute and about 500 Å/minute. In accordance with some embodiments, to increase the deposition rate of semiconductor layer **48B**, process conditions are adjusted. For example, the pressure of the deposition chamber, the flow rate (and/or the partial pressure) of the precursors (such as the silicon-containing precursor and/or the dopant-containing precursor), the wafer temperature, and/or the like may be increased.

[0053] In accordance with some embodiments, the partial pressure P2 of the silicon-containing gas in the formation of semiconductor layer **48B** is higher than the partial pressure P1 of the silicon-containing gas in the formation of semiconductor layer **48A**. For example, ratio P2/P1 is higher than 1.0, and may be in the range between about 1.1 and about 5 in accordance with some embodiments.

[0054] In accordance with some embodiments, the wafer temperature T2 in the formation of semiconductor layer **48B** is higher than the temperature T1 in the formation of semiconductor layer **48A**. For example, temperature difference (T2-T1) may be greater than about 25° C., and may be in the range between about 5° C. and about 250° C., or in the range between about 100° C. and about 250° C. in accordance with some embodiments.

[0055] In accordance with some embodiments, the flow rate FR2 of the silicon-containing gas in the formation of semiconductor layer **48B** is higher than the pressure flow rate FR1 of the silicon-containing gas in the formation of semiconductor layer **48A**. For example, ratio FR2/FR1 is higher than 1.0, and may be in the range between about 1 and about 5 in accordance with some embodiments.

[0056] In accordance with some embodiments, the growth (deposition) rate GR2 (the increase in thickness per unit time) of semiconductor layer **48B** is higher than the growth rate GR1 (during the deposition processes of the deposition-and-etch back cycles) of semiconductor layer **48A**. For example, the ratio GR2/GR1 is greater than 1.0, and may be in the range between about 2 and about 10 in accordance with some embodiments.

[0057] In accordance with some embodiments, to find the optimum ranges of process conditions for generating dislocations without causing other problems (such as the growth of semiconductor on dielectric materials), a plurality of sample wafers will be formed having the identical structures as in FIGS. **8B** and **15**. Source/drain regions **48** (including layers **48A** and **48B**) are grown in the sample wafers using different combinations of process conditions including, and not limited to, different growth rates, different wafer temperatures, different chamber pressures, and different flow rates. The resulting wafers are examined, for example, using Transmission Electron Microscopy

(TEM) to determine whether dislocations are formed or not, and to find the number of dislocations. The process conditions that will result in the desirable dislocations are used for the manufacturing of the wafers.

[0058] Dislocations **49** may include the dislocations **49A**, which starts to be formed when the growth of semiconductor layer **48B** is started. Accordingly, the beginning ends of dislocations **49A** may be at the interfaces of semiconductor layers **48A** and **48B**. Semiconductor layer **48A**, on the other hand, may not have dislocations therein. Alternatively, both of semiconductor layers **48A** and **48B** have dislocations **49** therein, while the number of dislocations in dielectric layer **48A** is significantly lower (for example, fewer than 5 percent) of the number of dislocations **49** in semiconductor layer **48B**. In accordance with these embodiments, some dislocations **49** starts to be formed in dielectric layer **48A**, while other dislocations **49** starts to be formed from the interfaces of semiconductor layers **48A** and **48B**.

[0059] Dislocations **49** may further include the dislocations **49B**, which have their beginning ends at the surfaces of dielectric features such as dielectric layer **46**, inner spacers **44**, gate spacers **38**, and the like. Each dislocation **49B** thus may have an end contacting a dielectric feature.

[0060] In accordance with alternative embodiments, rather than starting to adopt the process conditions for generating dislocations **49** (which process conditions include no etch-back process and higher growth rate), the process conditions for generating dislocations **49** may be adopted during the formation of semiconductor layer **48A**, or during the formation of semiconductor layer **48B**. For example, the formation of a lower part of semiconductor layer **48A** (or **48B**) may adopt etch-back process and/or lower wafer temperature, so that no dislocation is formed. The formation of an upper part of semiconductor layer **48A** (or **48B**), however, adopts different process conditions such as no etch-back process and higher wafer temperatures/flow rates, so that dislocations **49** starts to be formed when the upper part of semiconductor layer **48A** (or **48B**) is deposited.

[0061] In accordance with some embodiments, substrate **20** has a top surface orientation **[001]**, and a horizontal direction (for example, toward right) in directions. In accordance with some embodiments, dislocations **49** grow in the **[111]** directions. The tilt angle θ of dislocations **49** may be in the range between about 20 degrees and about 70 degrees, and may be about 54.7 degrees.

[0062] FIGS. **10A** and **10B** illustrate the cross-sectional views of the structure after the formation of Contact Etch Stop Layer (CESL) **50** and Inter-Layer Dielectric (ILD) **52**. The respective process is illustrated as process **218** in the process flow **200** shown in FIG. **21**. The corresponding structure is also shown in FIG. **18**. CESL **50** may be formed of silicon oxide, silicon nitride, silicon carbonitride, or the like, and may be formed using CVD, ALD, or the like. ILD **52** may include a dielectric material formed using, for example, FCVD, spin-on coating, CVD, or any other suitable deposition method. ILD **52** may be formed of an oxygen-containing dielectric material, which may be a silicon-oxide based material formed using Tetra Ethyl Ortho Silicate (TEOS) as a precursor, Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), Undoped Silicate Glass (USG), or the like.

[0063] CESL **50** and ILD **52** are planarized through a planarization process such as a CMP process or a mechanical grinding process. The respective process is illustrated as process **220** in the process flow **200** shown in FIG. **21**. In accordance with some embodiments, the planarization process may remove hard masks **36** to reveal dummy gate electrodes **34**, as shown in FIG. **10A**. In accordance with alternative embodiments, the planarization process may reveal, and is stopped on, hard masks **36**. In accordance with some embodiments, after the planarization process, the top surfaces of dummy gate electrodes **34** (or hard masks **36**), gate spacers **38**, and ILD **52** are level within process variations.

[0064] Next, dummy gate electrodes **34** and dummy gate dielectrics **32** (and hard masks **36**, if remaining) are removed in one or more etching processes, so that recesses **58** are formed, as shown in FIGS. **11A** and **11B**. The respective process is illustrated as process **222** in the process flow **200** shown in FIG. **21**. In accordance with some embodiments, dummy gate electrodes **34** and dummy

gate dielectrics **32** are removed through an anisotropic dry etch process(es). For example, the etching process may be performed using reaction gas(es) that selectively etch dummy gate electrodes **34** and dummy gate dielectrics **32** at faster rates than ILD **52**. Each recess **58** exposes and/or overlies portions of multilayer stacks **22'**, which include the future channel regions in subsequently completed transistors.

[0065] Sacrificial layers **22A** are then removed to extend recesses **58** between nanostructures **22B**. The respective process is illustrated as process **224** in the process flow **200** shown in FIG. **21**.

Sacrificial layers **22A** may be removed by performing an isotropic etching process such as a wet etching process using etchants which are selective to the materials of sacrificial layers **22A**, while nanostructures **22B**, substrate **20**, and STI regions **26** remain relatively un-etched as compared to sacrificial layers **22A**. In accordance with some embodiments in which sacrificial layers **22A** include, for example, SiGe, and nanostructures **22B** include, for example, Si or SiC, tetra methyl ammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH), or the like may be used to remove sacrificial layers **22A**.

[0066] Referring to FIGS. **12A** and **12B**, gate dielectrics **62** and gate electrodes **68** are formed, hence forming replacement gate stacks **70**. The respective process is illustrated as process **226** in the process flow **200** shown in FIG. **21**. The corresponding structure is also shown in FIG. **19**. In accordance with some embodiments, each of gate dielectric **62** includes an interfacial layer and a high-k dielectric layer on the interfacial layer. The interfacial layer may be formed of or comprises silicon oxide, which may be deposited through a conformal deposition process such as ALD or CVD, or through an oxidation process. In accordance with some embodiments, the high-k dielectric layers comprise one or more dielectric layers. For example, the high-k dielectric layer(s) may include a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof.

[0067] Gate electrodes **68** are also formed. In the formation, conductive layers are first formed on the high-k dielectric layer, and the remaining portions of recesses **58** are filled. Gate electrodes **68** may include a metal-containing material such as TiN, TaN, TiAl, TiAlC, cobalt, ruthenium, aluminum, tungsten, combinations thereof, and/or multilayers thereof. For example, gate electrodes **68** may comprise any number of layers, any number of work function layers, and possibly a filling material. Gate dielectrics **62** and gate electrodes **68** also fill the spaces between adjacent ones of nanostructures **22B**, and fill the spaces between the bottom ones of nanostructures **22B** and the underlying substrate strips **20'**. After the filling of recesses **58**, a planarization process such as a CMP process or a mechanical grinding process is performed to remove the excess portions of the gate dielectrics and the material of gate electrodes **68**, which excess portions are over the top surface of ILD **52**. Gate electrodes **68** and gate dielectrics **62** are collectively referred to as gate stacks **70** of the resulting transistors.

[0068] In the processes shown in FIGS. **13A** and **13B**, gate stacks **70** are recessed, so that recesses are formed directly over gate stacks **70** and between opposing portions of gate spacers **38**. A gate mask **74** comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in each of the recesses, followed by a planarization process to remove excess portions of the dielectric material extending over ILD **52**. The respective process is illustrated as process **228** in the process flow **200** shown in FIG. **21**.

[0069] As further illustrated by FIGS. **13A** and **13B**, ILD **76** is deposited over ILD **52** and over gate masks **74**. The respective process is illustrated as process **230** in the process flow **200** shown in FIG. **21**. An etch stop layer (not shown) may be (or may not be) deposited before the formation of ILD **76**. In accordance with some embodiments, ILD **76** is formed through FCVD, CVD, PECVD, or the like. ILD **76** is formed of a dielectric material, which may be selected from silicon oxide, PSG, BSG, BPSG, USG, or the like.

[0070] In FIGS. **14A** and **14B**, ILD **76**, ILD **52**, CESL **50**, and gate masks **74** are etched to form recesses (occupied by contact plugs **80A** and **80B**) exposing surfaces of source/drain regions **48**

and/or gate stacks **70**. The recesses may be formed through etching using an anisotropic etching process, such as RIE, NBE, or the like. Although FIG. **14B** illustrates that contact plugs **80A** and **80B** are in a same cross-section, in various embodiments, contact plugs **80A** and **80B** may be formed in different cross-sections, thereby reducing the risk of shorting with each other.

[0071] After the recesses are formed, silicide regions **78** are formed over source/drain regions **48**. The respective process is illustrated as process **232** in the process flow **200** shown in FIG. **21**. Contact plugs **80B** are then formed over silicide regions **78**. Also, contacts **80A** (may also be referred to as gate contact plugs) are also formed in the recesses, and are over and contacting gate electrodes **68**. The respective process is illustrated as process **234** in the process flow **200** shown in FIG. **21**. The corresponding structure is also shown in FIG. **20A**. Transistor **82** is thus formed. It is noted that the details of the source/drain regions **48** and dislocations **49** are not shown in FIGS. **14A** and **14B**, and the details may be found referring to FIG. **20A**.

[0072] Due to the formation of dislocations **49**, the metals used in the processes following the formation of source/drain regions **48** have a higher chance to diffuse to and through dislocations **49**, thus metal ions have higher concentrations at dislocations **49** than in the parts of source/drain regions **48** neighboring the dislocations **49**. For example, FIG. **20A** schematically illustrates the concentrated metal ions **84** along and at dislocations **49**, which metal ions **84** may include the ions of an alkali metal(s) (such as lithium, sodium, potassium), tungsten, cobalt, nickel, titanium, tantalum, or the like. Accordingly, the concentrated metal ions **84** have concentrations higher than the parts of the source/drain regions **48** apart from the dislocations **49**. The diffusion of the metal ions **84** may occur during the subsequent formation (such as the deposition and the CMP) of contact plugs **80B**. The concentration of the metal ions **84** may be observed using elemental analysis such as Electron Dispersive X-ray Spectroscopy (EDX) or Atom Probe Tomography (APT).

[0073] FIG. **20B** illustrates parts of GAA transistor **82'** in accordance with some embodiments. GAA transistor **82'** may have essentially the same structure as, and formed using essentially the same processes as, the GAA transistor **82** in FIG. **20A**. GAA transistors **82'** and **82** may be formed in the same device die and on the same semiconductor substrate **20**. The source/drain region **48** of GAA transistor **82'** is free from source/drain regions, so that GAA transistor **82'** has lower drive current than GAA transistor **82** to suit to customized design requirement.

[0074] In accordance with some embodiments, most of the formation processes of GAA transistor **82'** may be shared with GAA transistor **82**, except for the formation of source/drain regions **48**. The formation of the source/drain regions **48** of GAA transistor **82** is tuned for forming dislocations, while the formation of the source/drain regions **48** of GAA transistor **82'** is tuned to be free from dislocations. In accordance with some embodiments, the semiconductor layers **48A** of GAA transistors **82** and **82'** share a common formation process(es), while the semiconductor layers **48B** of GAA transistors **82** and **82'** are formed by separate processes, so that GAA transistor **82** has dislocations **49**, and the GAA transistor **82'** are free from dislocations.

[0075] The embodiments of the present disclosure have some advantageous features. By adjusting process conditions, dislocations may be formed in source/drain regions. The dislocations result in the increased stress to the channel regions. The currents of the resulting GAA transistors are thus increased.

[0076] In accordance with some embodiments of the present disclosure, a method comprises forming a protruding feature comprising a first sacrificial nanosheet over a bulk semiconductor substrate; a first semiconductor nanosheet over the first sacrificial nanosheet; a second sacrificial nanosheet over the first semiconductor nanosheet; and a second semiconductor nanosheet over the second sacrificial nanosheet; forming a dummy gate stack on the protruding feature; etching the protruding feature to form a recess; forming a source/drain region in the recess, wherein dislocations are formed in the source/drain region; removing the first sacrificial nanosheet and the second sacrificial nanosheet; and forming a replacement gate stack to replace the dummy gate

stack.

[0077] In an embodiment, the method further comprises, before the source/drain region is formed in the recess, forming a dielectric layer at a bottom of the recess. In an embodiment, some of the dislocations are formed starting from the dielectric layer. In an embodiment, the forming the source/drain region comprises epitaxially growing a first semiconductor layer; and epitaxially growing a second semiconductor layer different from the first semiconductor layer, wherein the dislocations start to grow when the second semiconductor layer is grown.

[0078] In an embodiment, the growing the first semiconductor layer comprises a plurality of cycles, each comprising depositing a layer of the first semiconductor layer; and etching back the layer of the first semiconductor layer, and wherein the growing the second semiconductor layer is a continuous process that ends after the second semiconductor layer has a first top surface higher than a second top surface of the second semiconductor nanosheet. In an embodiment, the growing the second semiconductor layer is performed without etch-back process therein. In an embodiment, the growing the first semiconductor layer is performed at a first wafer temperature, and the growing the second semiconductor layer is performed at a second wafer temperature higher than the first wafer temperature.

[0079] In an embodiment, the growing the first semiconductor layer is performed with a first flow rate of a silicon-containing precursor, and the growing the second semiconductor layer is performed with a second flow rate of the silicon-containing precursor, and wherein the second flow rate is higher than the first flow rate. In an embodiment, the growing the first semiconductor layer is performed with a first partial pressure of a silicon-containing precursor, and the growing the second semiconductor layer is performed with a second partial pressure of the silicon-containing precursor, and the second partial pressure is higher than the first partial pressure. In an embodiment, all of the dislocations in the source/drain region are spaced apart from all semiconductor nanosheets in the protruding feature.

[0080] In accordance with some embodiments of the present disclosure, a device comprises a first semiconductor nanostructure; a second semiconductor nanostructure over the first semiconductor nanostructure; a gate stack comprising a portion between the first semiconductor nanostructure and the second semiconductor nanostructure; a source/drain region aside of and joined to the first semiconductor nanostructure and the second semiconductor nanostructure, wherein the first semiconductor nanostructure, the second semiconductor nanostructure, the gate stack, and the source/drain region form parts of a transistor; and a first dislocation in the source/drain region.

[0081] In an embodiment, the device further comprises a second dislocation in the source/drain region and parallel to the first dislocation. In an embodiment, the device further comprises metal ions concentrated at the first dislocation, wherein the metal ions have a higher metal ion concentration at the first dislocation than surrounding parts of the source/drain region. In an embodiment, the device further comprises a dielectric layer underlying and contacting the source/drain region, wherein the first dislocation has an end contacting the dielectric layer.

[0082] In an embodiment, the first dislocation is spaced apart from all semiconductor nanostructures in the transistor. In an embodiment, the source/drain region comprises a first semiconductor layer contacting the first semiconductor nanostructure; and a second semiconductor layer different from the first semiconductor layer, wherein an end of the first dislocation is at an interface between the first semiconductor layer and the second semiconductor layer. In an embodiment, the device further comprises an inner spacer contacting the portion of the gate stack, wherein the first dislocation has an end contacting the inner spacer.

[0083] In accordance with some embodiments of the present disclosure, a device comprises a first plurality of semiconductor nanostructures, wherein upper ones of the first plurality of semiconductor nanostructures overlap respective lower ones of the first plurality of semiconductor nanostructures; a first gate stack comprising portions between the first plurality of semiconductor nanostructures; a second plurality of semiconductor nanostructures, wherein upper ones of the

second plurality of semiconductor nanostructures overlap respective lower ones of the second plurality of semiconductor nanostructures; a second gate stack comprising portions between the second plurality of semiconductor nanostructures; a source/drain region between the first plurality of semiconductor nanostructures and the second plurality of semiconductor nanostructures; a first plurality of dislocations in the source/drain region and parallel to each other, wherein the first plurality of dislocations comprise first lower ends close to the first plurality of semiconductor nanostructures; and a second plurality of dislocations in the source/drain region and parallel to each other, wherein the second plurality of dislocations comprise second lower ends close to the second plurality of semiconductor nanostructures.

[0084] In an embodiment, the first plurality of dislocations are spaced apart from the first plurality of semiconductor nanostructures by a portion of the source/drain region. In an embodiment, the portion of the source/drain region separating the first plurality of dislocations apart from the first plurality of semiconductor nanostructures has a different composition than the portions of the source/drain region comprising the dislocations therein.

[0085] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming a protruding feature comprising: a first sacrificial nanosheet over a bulk semiconductor substrate; a first semiconductor nanosheet over the first sacrificial nanosheet; a second sacrificial nanosheet over the first semiconductor nanosheet; and a second semiconductor nanosheet over the second sacrificial nanosheet; forming a dummy gate stack on the protruding feature; etching the protruding feature to form a recess; forming a source/drain region in the recess, wherein dislocations are formed in the source/drain region; removing the first sacrificial nanosheet and the second sacrificial nanosheet; and forming a replacement gate stack to replace the dummy gate stack.
2. The method of claim 1 further comprising, before the source/drain region is formed in the recess, forming a dielectric layer at a bottom of the recess.
3. The method of claim 2, wherein some of the dislocations are formed starting from the dielectric layer.
4. The method of claim 1, wherein the forming the source/drain region comprises: epitaxially growing a first semiconductor layer; and epitaxially growing a second semiconductor layer different from the first semiconductor layer, wherein the dislocations start to grow when the second semiconductor layer is grown.
5. The method of claim 4, wherein the growing the first semiconductor layer comprises a plurality of cycles, each comprising: depositing a layer of the first semiconductor layer; and etching back the layer of the first semiconductor layer, and wherein the growing the second semiconductor layer is a continuous process that ends after the second semiconductor layer has a first top surface higher than a second top surface of the second semiconductor nanosheet.
6. The method of claim 5, wherein the growing the second semiconductor layer is performed without etch-back process therein.
7. The method of claim 4, wherein the growing the first semiconductor layer is performed at a first wafer temperature, and the growing the second semiconductor layer is performed at a second wafer

temperature higher than the first wafer temperature.

8. The method of claim 4, wherein the growing the first semiconductor layer is performed with a first flow rate of a silicon-containing precursor, and the growing the second semiconductor layer is performed with a second flow rate of the silicon-containing precursor, and wherein the second flow rate is higher than the first flow rate.

9. The method of claim 4, wherein the growing the first semiconductor layer is performed with a first partial pressure of a silicon-containing precursor, and the growing the second semiconductor layer is performed with a second partial pressure of the silicon-containing precursor, and the second partial pressure is higher than the first partial pressure.

10. The method of claim 1, wherein all of the dislocations in the source/drain region are spaced apart from all semiconductor nanosheets in the protruding feature.

11. A device comprising: a first semiconductor nanostructure; a second semiconductor nanostructure over the first semiconductor nanostructure; a gate stack comprising a portion between the first semiconductor nanostructure and the second semiconductor nanostructure; a source/drain region aside of and joined to the first semiconductor nanostructure and the second semiconductor nanostructure, wherein the first semiconductor nanostructure, the second semiconductor nanostructure, the gate stack, and the source/drain region form parts of a transistor; and a first dislocation in the source/drain region.

12. The device of claim 11 further comprising a second dislocation in the source/drain region and parallel to the first dislocation.

13. The device of claim 11 further comprising metal ions concentrated at the first dislocation, wherein the metal ions have a higher metal ion concentration at the first dislocation than surrounding parts of the source/drain region.

14. The device of claim 11 further comprising a dielectric layer underlying and contacting the source/drain region, wherein the first dislocation has an end contacting the dielectric layer.

15. The device of claim 11, wherein the first dislocation is spaced apart from all semiconductor nanostructures in the transistor.

16. The device of claim 11, wherein the source/drain region comprises: a first semiconductor layer contacting the first semiconductor nanostructure; and a second semiconductor layer different from the first semiconductor layer, wherein an end of the first dislocation is at an interface between the first semiconductor layer and the second semiconductor layer.

17. The device of claim 11 further comprising an inner spacer contacting the portion of the gate stack, wherein the first dislocation has an end contacting the inner spacer.

18. A device comprising: a first plurality of semiconductor nanostructures, wherein upper ones of the first plurality of semiconductor nanostructures overlap respective lower ones of the first plurality of semiconductor nanostructures; a first gate stack comprising portions between the first plurality of semiconductor nanostructures; a second plurality of semiconductor nanostructures, wherein upper ones of the second plurality of semiconductor nanostructures overlap respective lower ones of the second plurality of semiconductor nanostructures; a second gate stack comprising portions between the second plurality of semiconductor nanostructures; a source/drain region between the first plurality of semiconductor nanostructures and the second plurality of semiconductor nanostructures; a first plurality of dislocations in the source/drain region and parallel to each other, wherein the first plurality of dislocations comprise first lower ends close to the first plurality of semiconductor nanostructures; and a second plurality of dislocations in the source/drain region and parallel to each other, wherein the second plurality of dislocations comprise second lower ends close to the second plurality of semiconductor nanostructures.

19. The device of claim 18, wherein the first plurality of dislocations are spaced apart from the first plurality of semiconductor nanostructures by a portion of the source/drain region.

20. The device of claim 19, wherein the portion of the source/drain region separating the first plurality of dislocations apart from the first plurality of semiconductor nanostructures has a

different composition than the portions of the source/drain region comprising the dislocations therein.
